PHASE 2

- Read the complete datasheet for the LT part and Demo Board.
- Run detailed simulations of your converter.
- Make sure to study waveforms specific/unique to your converter topology.
- Capture relevant screen shots to demonstrate function.
- Study your evaluation board and finalize measurement setup for your converter.

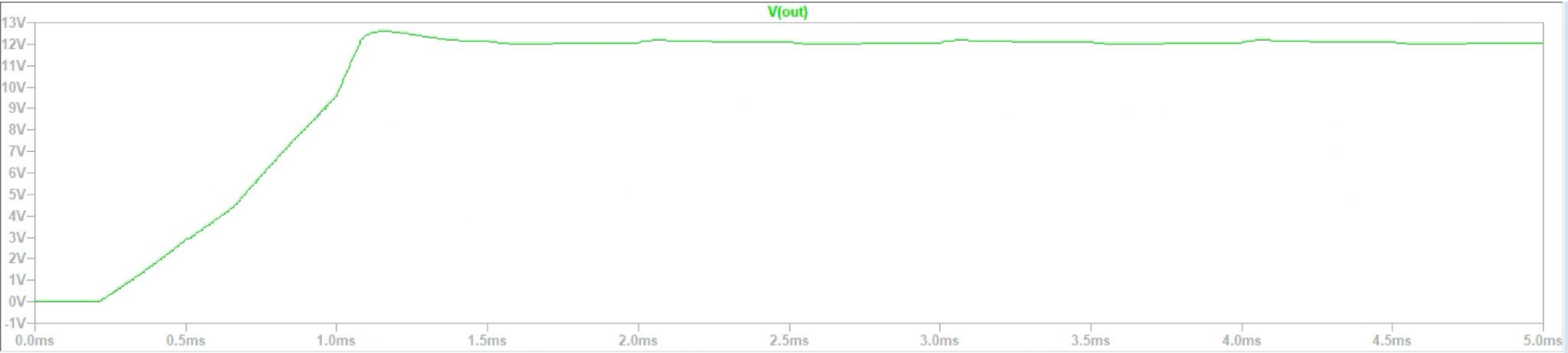
Make sure you know what is needed and you have all the necessary instruments.

- Prepare slides for the below bullets

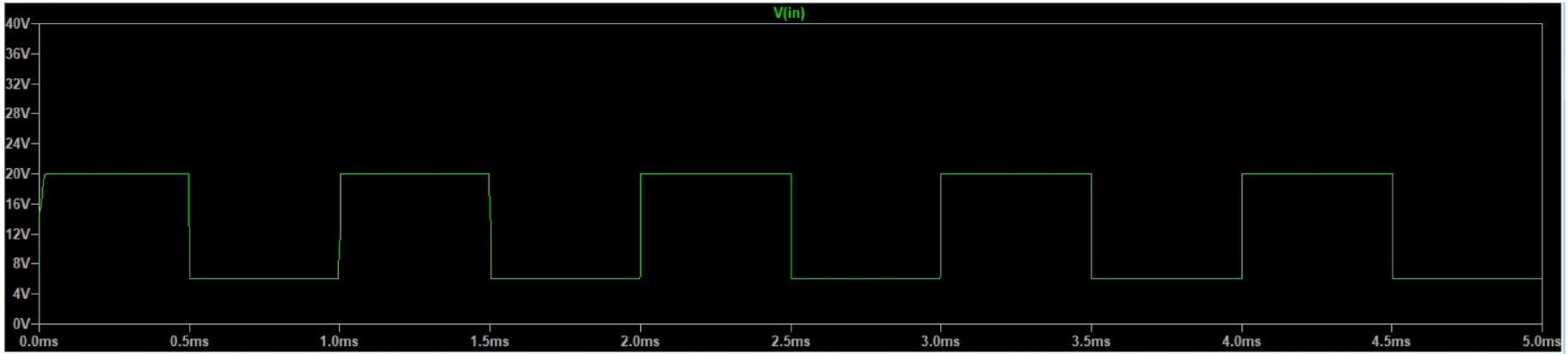
Simulation setup and relevant schematic

Simulation snap shots that show functional operation.

V(out)

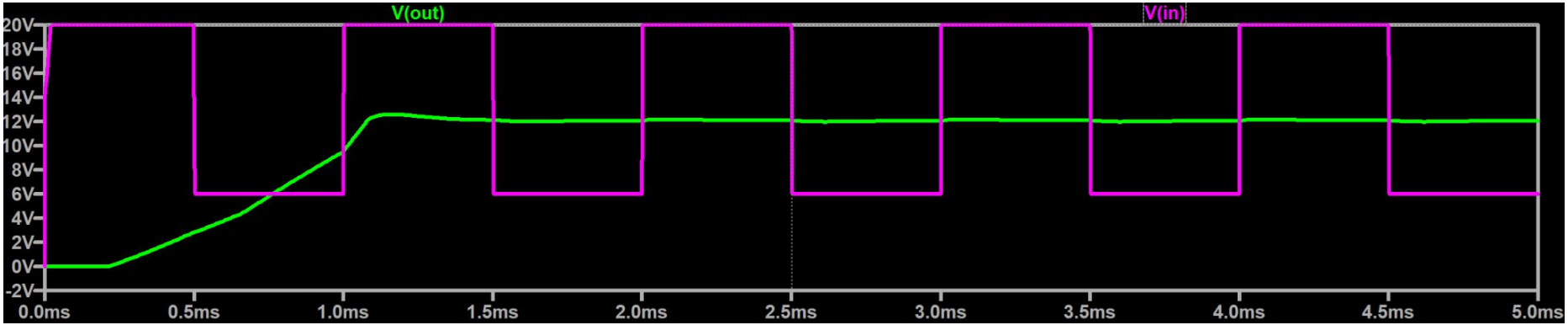


V(in) → data sheet says max 36V, graph shows max 20V

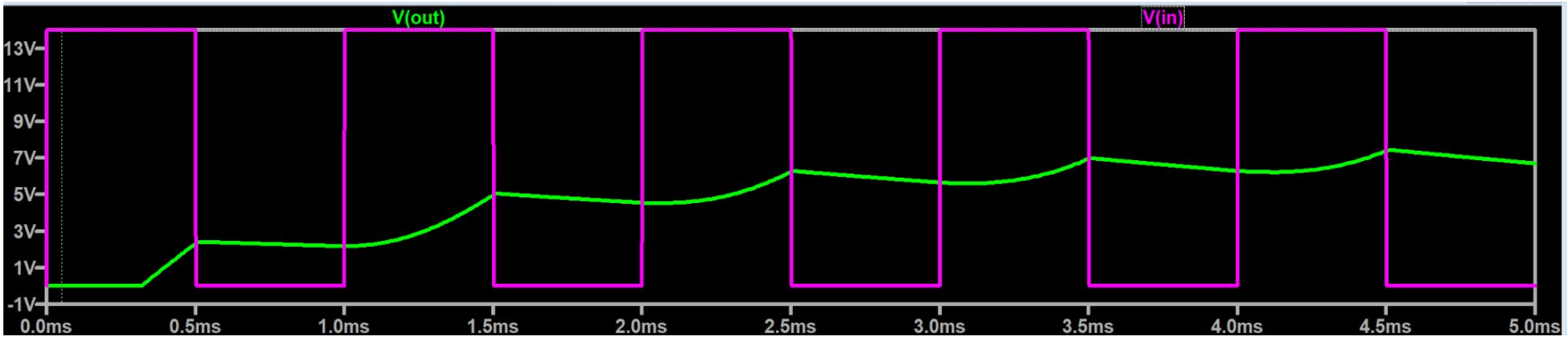


Changing DC Voltage – Change value of V2

DC Offset 6V (default)

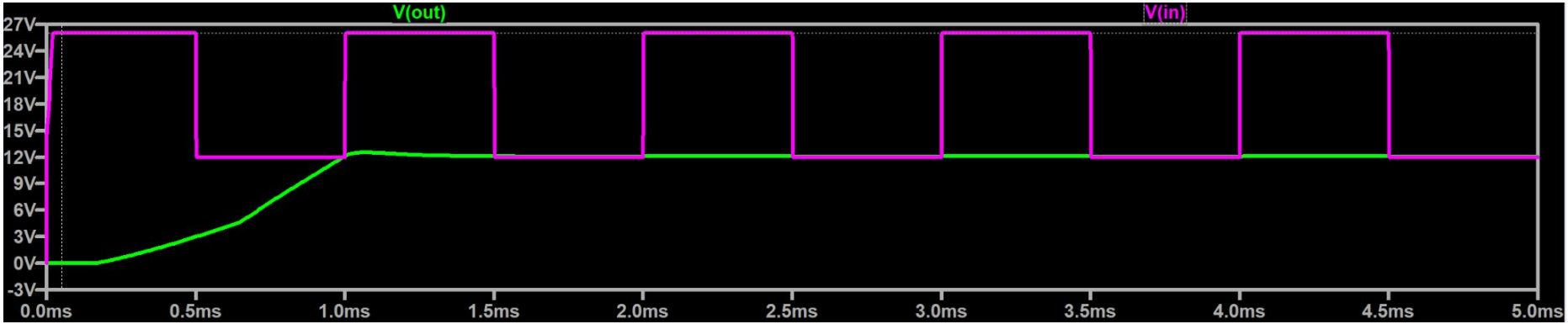


DC Offset 0V



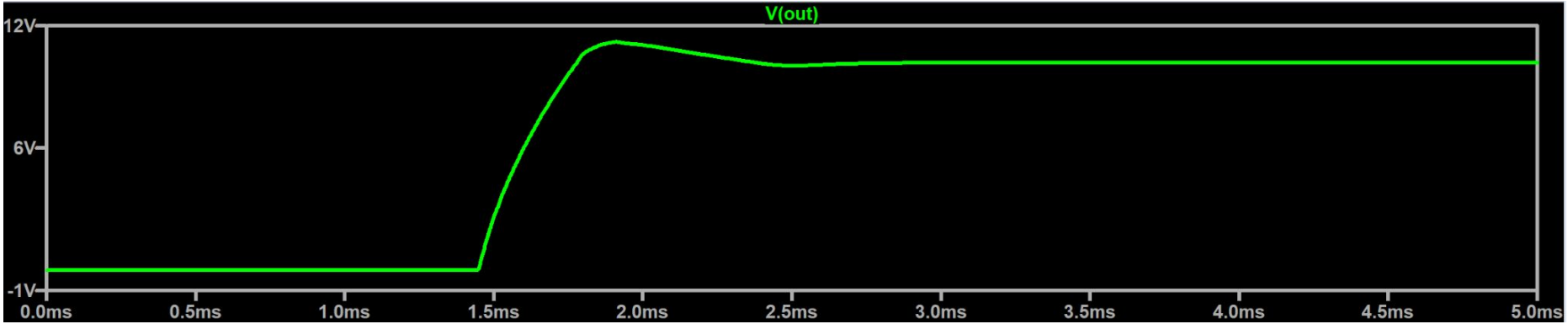
Changing DC Voltage Cont.

DC Offset 12V

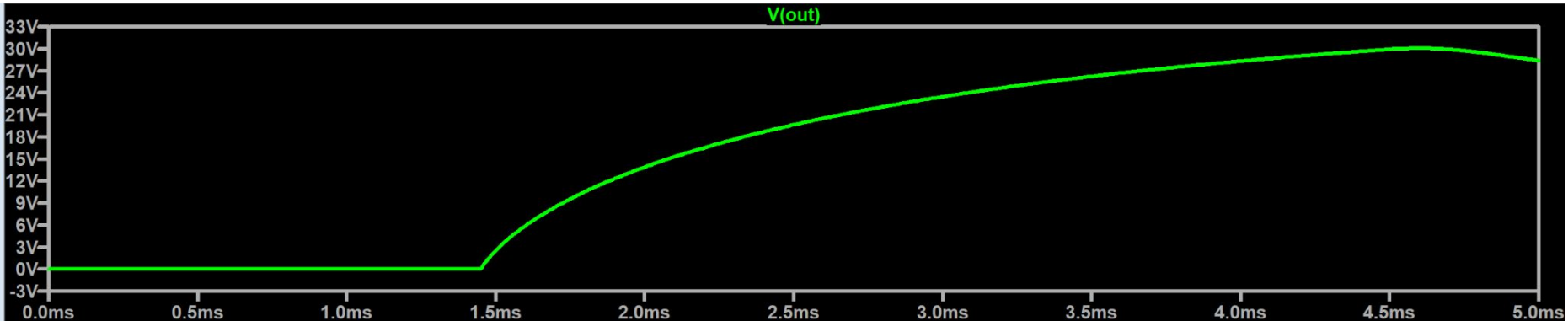


Change R4 (bottom feedback)

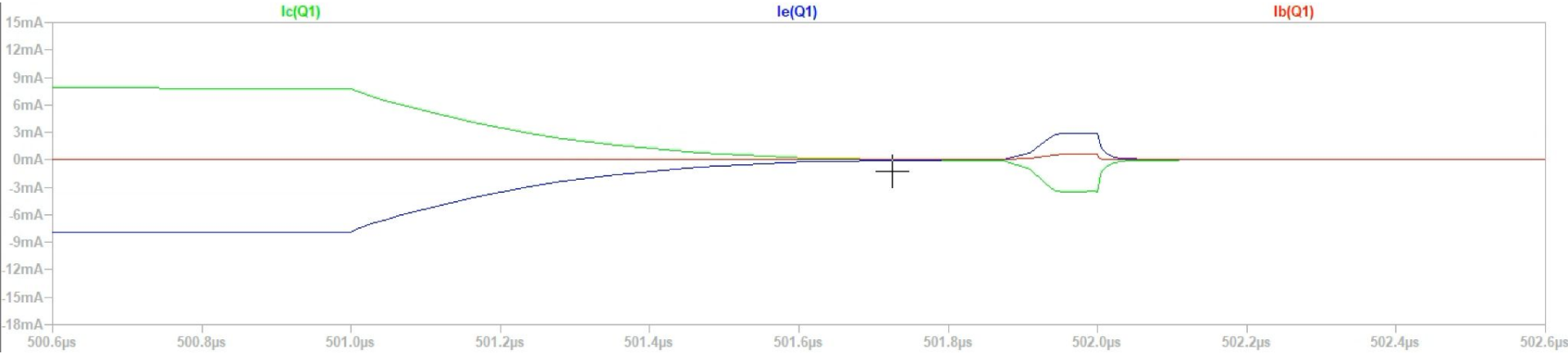
R4 = 15kOhms, Vout = 10.15V



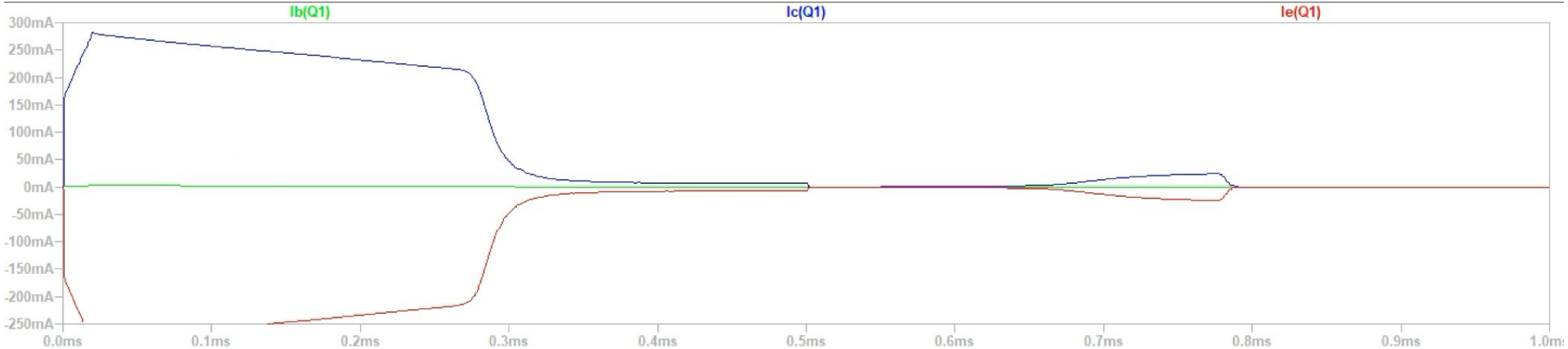
R4 = 5kOhms, Vout = 29.98V



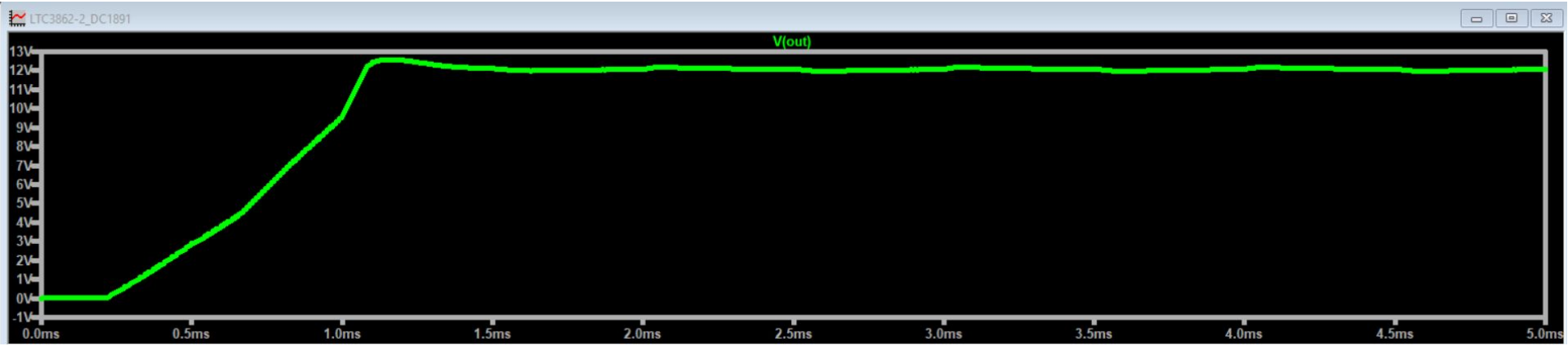
Base, Emitter and collector currents with V2=6V:

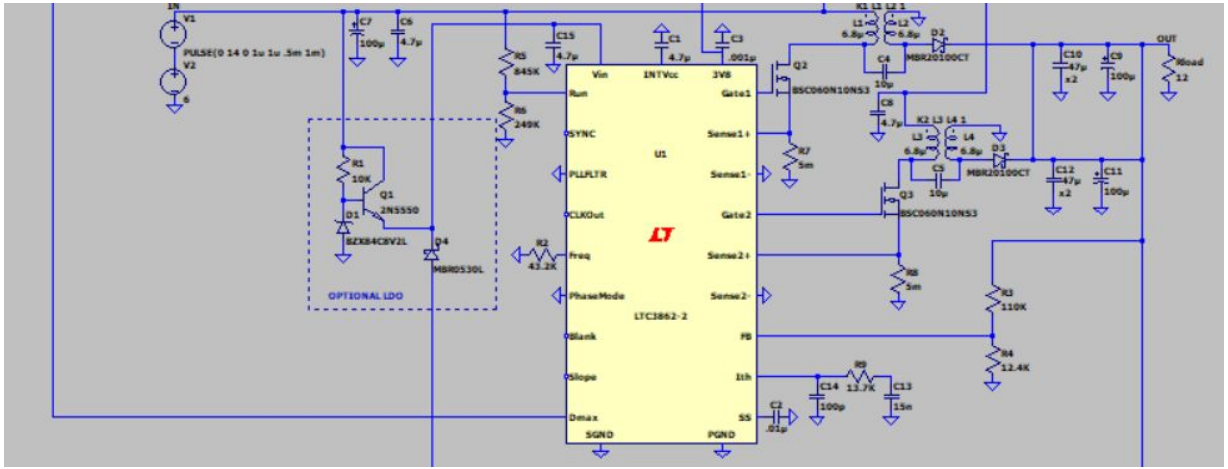


Base, Emitter and collector currents with V2=12V:

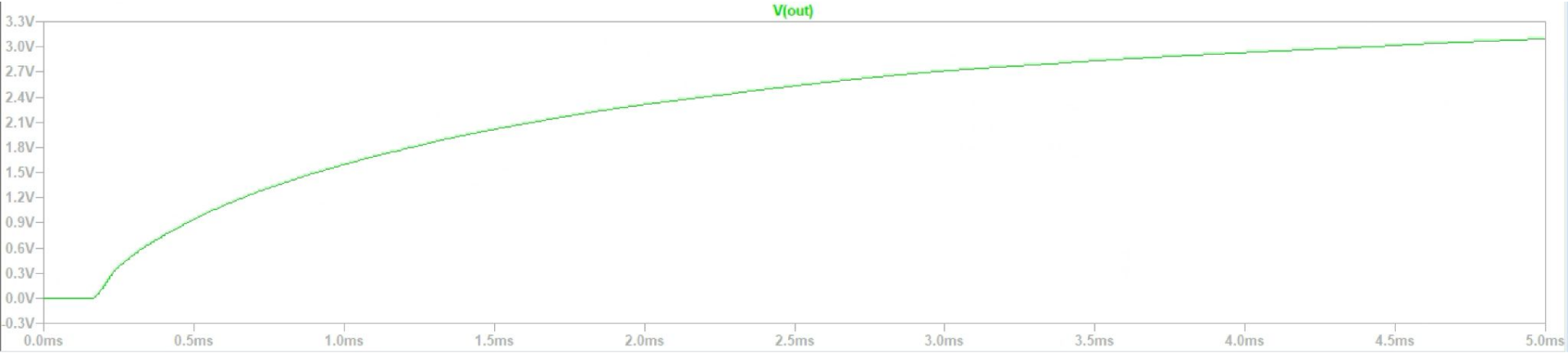


Setting max Duty Cycle to 75%

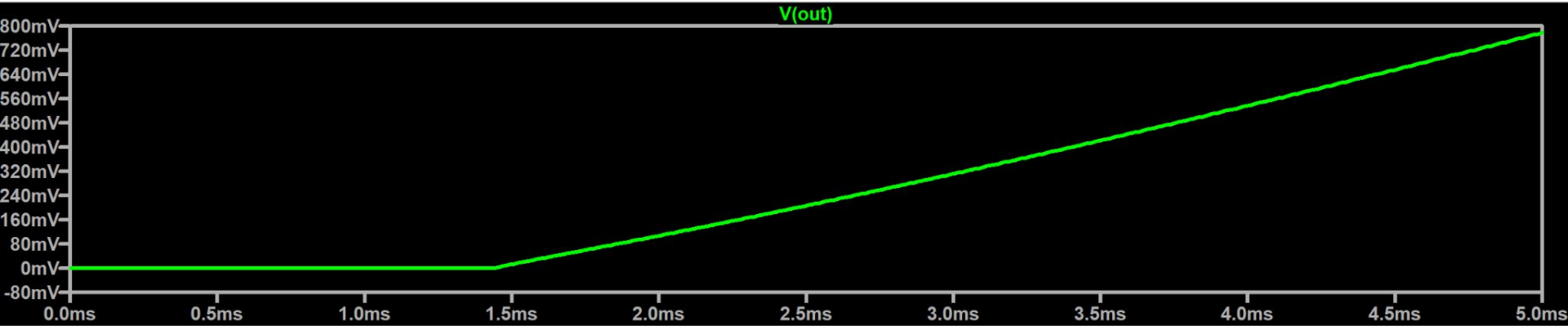




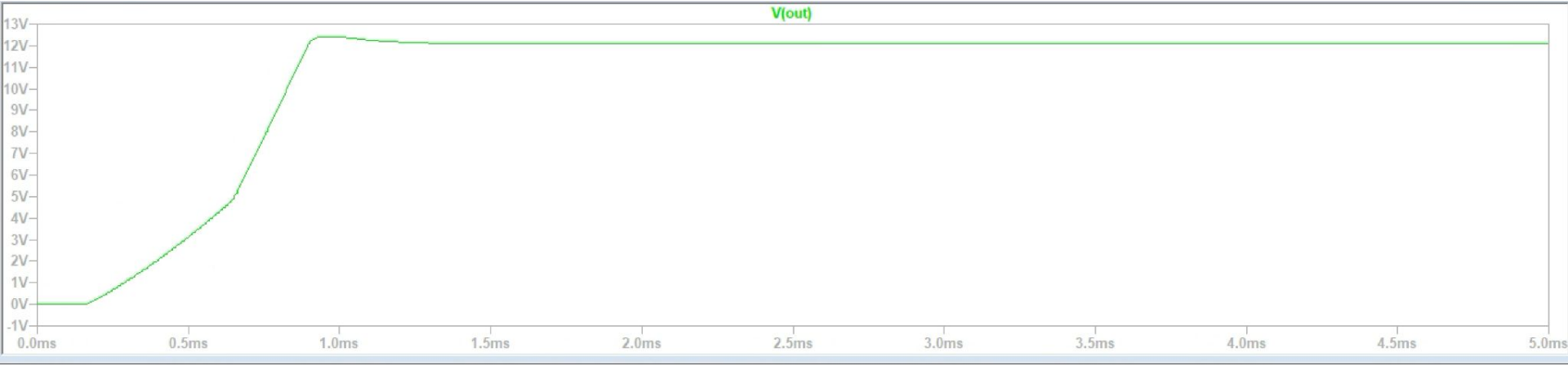
SS with no 0.01μ Capacitor:

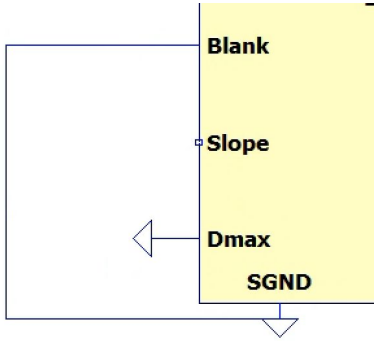


Change SS to 1μ Capacitor:



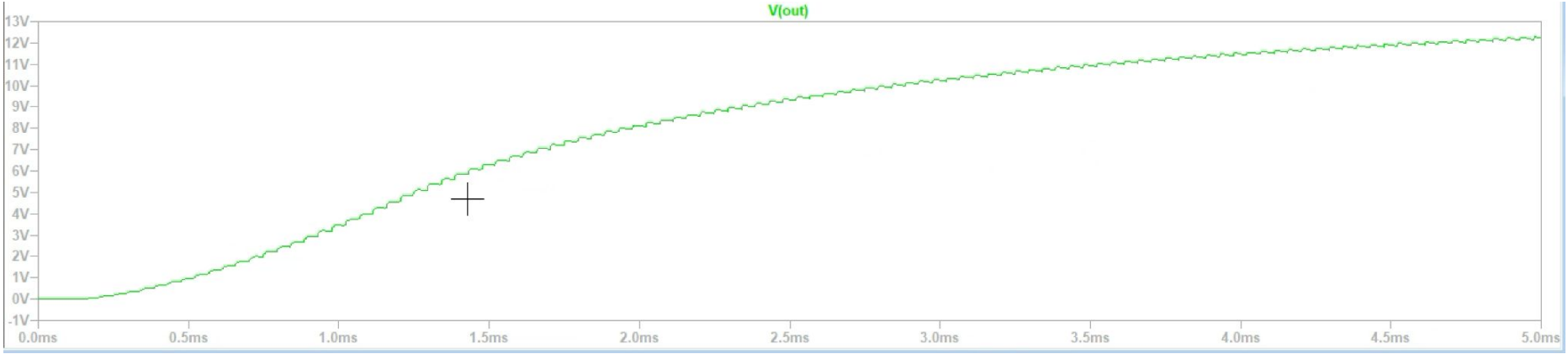
Connecting BLANK to SGND

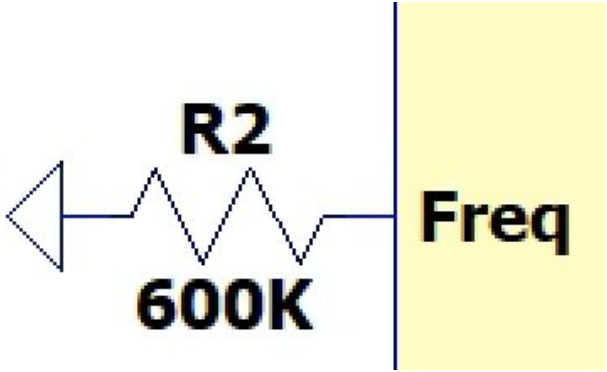


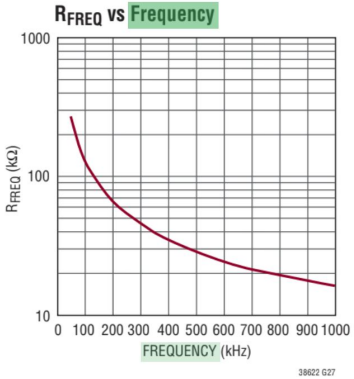


BLANK (Pin 3/Pin 1/Pin 3): Blanking Time. Floating this pin provides a nominal minimum on-time of 290ns. Connecting this pin to 3V8 provides a minimum on-time of 375ns, while connecting it to SGND provides a minimum on-time of 210ns.

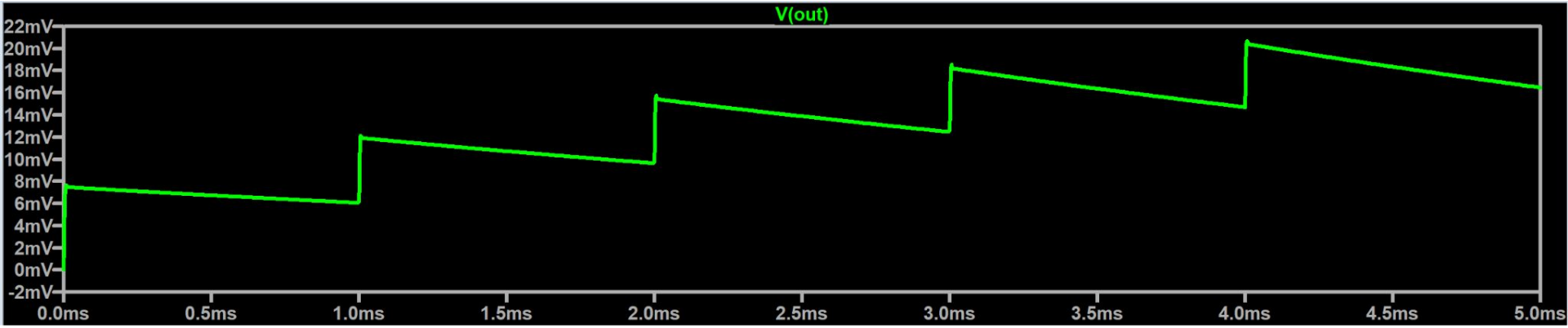
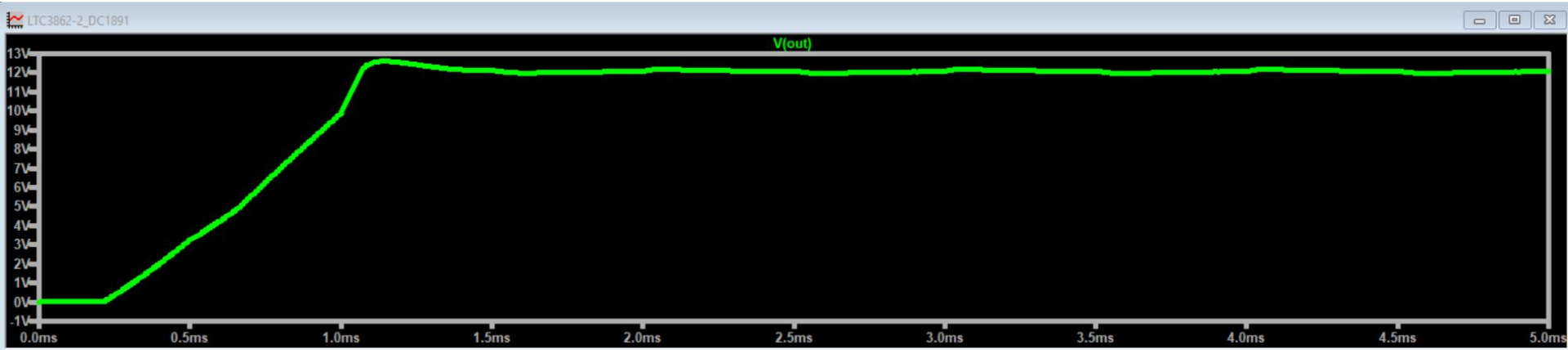
When external resistor is over max







When Freq Resistor is Under Min → R_{freq} = 100ohm



Changing Rfreq to 35kOhm (still in range)